

FIN302 Tutorial 11 Answers

Q 13.11

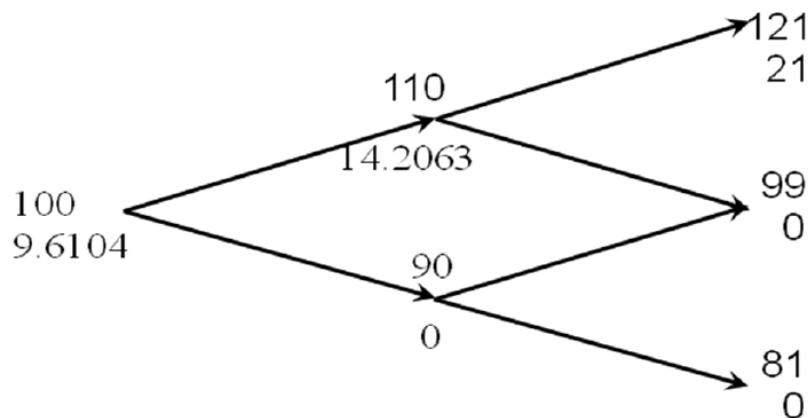
In this case, $u = 1.10$, $d = 0.90$, $\Delta t = 0.5$, and $r = 0.08$, so that

$$p = \frac{e^{0.08 \times 0.5} - 0.90}{1.10 - 0.90} = 0.7041$$

The tree for stock price movements is shown in Figure S13.1. We can work back from the end of the tree to the beginning, as indicated in the diagram, to give the value of the option as \$9.61. The option value can also be calculated directly from equation (13.10):

$$[0.7041^2 \times 21 + 2 \times 0.7041 \times 0.2959 \times 0 + 0.2959^2 \times 0]e^{-2 \times 0.08 \times 0.5} = 9.61$$

or \$9.61.



Q13.12

Figure S13.2 shows how we can value the put option using the same tree as in Problem 13.1. The value of the option is \$1.92. The option value can also be calculated directly from equation (13.10):

$$e^{-2 \times 0.08 \times 0.5} [0.7041^2 \times 0 + 2 \times 0.7041 \times 0.2959 \times 1 + 0.2959^2 \times 19] = 1.92$$

or \$1.92. The stock price plus the put price is $100 + 1.92 = \$101.92$. The present value of the

strike price plus the call price is $100e^{-0.08 \times 1} + 9.61 = \101.92 . These are the same, verifying that put-call parity holds.

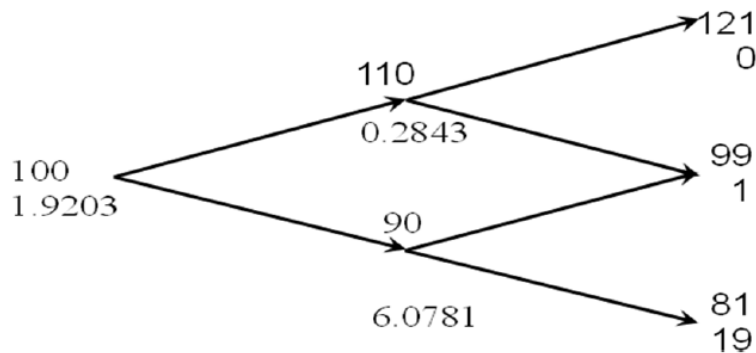


Figure S13.2: Tree for Problem 13.2

Q13.14

At the end of two months, the value of the option will be either \$4 (if the stock price is \$53) or \$0 (if the stock price is \$48). Consider a portfolio consisting of:

+ Δ : shares

-1 : option

The value of the portfolio is either 48Δ or $53\Delta - 4$ in two months. If

$$48\Delta = 53\Delta - 4$$

that is,

$$\Delta = 0.8$$

the value of the portfolio is certain to be 38.4. For this value of Δ , the portfolio is therefore riskless. The current value of the portfolio is:

$$0.8 \times 50 - f$$

where f is the value of the option. Since the portfolio must earn the risk-free rate of interest

$$(0.8 \times 50 - f)e^{0.10 \times 2/12} = 38.4$$

that is

$$f = 2.23$$

The value of the option is therefore \$2.23.

This can also be calculated directly from equations (13.2) and (13.3). $u = 1.06$, $d = 0.96$ so that

$$p = \frac{e^{0.10 \times 2/12} - 0.96}{1.06 - 0.96} = 0.5681$$

and

$$f = e^{-0.10 \times 2/12} \times 0.5681 \times 4 = 2.23$$

Q13.16

At the end of three months the value of the option is either \$5 (if the stock price is \$35) or \$0 (if the stock price is \$45).

Consider a portfolio consisting of:

$-\Delta$: shares

$+1$: option

(Note: The delta, Δ , of a put option is negative. We have constructed the portfolio so that it is $+1$ option and $-\Delta$ shares rather than -1 option and $+\Delta$ shares so that the initial investment is positive.)

The value of the portfolio is either $-35\Delta + 5$ or -45Δ . If:

$$-35\Delta + 5 = -45\Delta$$

that is,

$$\Delta = -0.5$$

the value of the portfolio is certain to be 22.5. For this value of Δ , the portfolio is therefore riskless. The current value of the portfolio is

$$-40\Delta + f$$

where f is the value of the option. Since the portfolio must earn the risk-free rate of interest

$$(40 \times 0.5 + f) \times 1.02 = 22.5$$

Hence,

$$f = 2.06$$

i.e., the value of the option is \$2.06.

This can also be calculated using risk-neutral valuation. Suppose that p is the probability of an upward stock price movement in a risk-neutral world. We must have

$$45p + 35(1 - p) = 40 \times 1.02$$

that is

$$10p = 5.8$$

Or,

$$p = 0.58$$

The expected value of the option in a risk-neutral world is:

$$0 \times 0.58 + 5 \times 0.42 = 2.10$$

This has a present value of

$$\frac{2.10}{1.02} = 2.06$$

This is consistent with the no-arbitrage answer.

Q13.17.

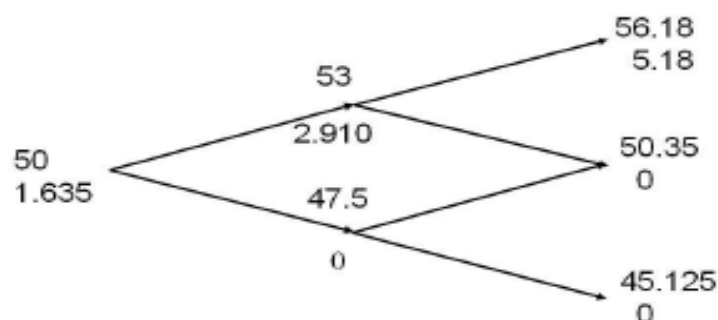
A tree describing the behavior of the stock price is shown in Figure S13.3. The risk-neutral probability of an up move, p , is given by

$$p = \frac{e^{0.05 \times 3/12} - 0.95}{1.06 - 0.95} = 0.5689$$

There is a payoff from the option of $56.18 - 51 = 5.18$ for the highest final node (which corresponds to two up moves) zero in all other cases. The value of the option is therefore

$$5.18 \times 0.5689^2 \times e^{-0.05 \times 6/12} = 1.635$$

This can also be calculated by working back through the tree as indicated in Figure S13.3. The value of the call option is the lower number at each node in the figure.



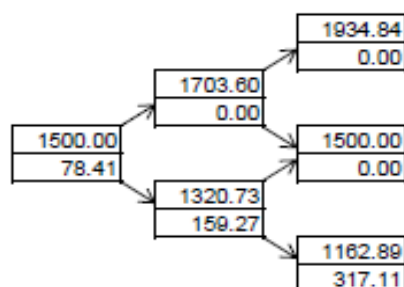
Q13.22

$$u = e^{0.18 \times \sqrt{0.5}} = 1.1357$$

$$d = 1/u = 0.8805$$

$$p = \frac{e^{(0.04 - 0.025) \times 0.5} - 0.8805}{1.1357 - 0.8805} = 0.4977$$

The tree is shown in Figure S13.6. The option is exercised at the lower node at the six-month point. It is worth 78.41.



Q15.13

In this case, $S_0 = 50$, $K = 50$, $r = 0.1$, $\sigma = 0.3$, $T = 0.25$, and

$$d_1 = \frac{\ln(50/50) + (0.1 + 0.09/2)0.25}{0.3\sqrt{0.25}} = 0.2417$$

$$d_2 = d_1 - 0.3\sqrt{0.25} = 0.0917$$

The European put price is

$$\begin{aligned} & 50N(-0.0917)e^{-0.1 \times 0.25} - 50N(-0.2417) \\ &= 50 \times 0.4634e^{-0.1 \times 0.25} - 50 \times 0.4045 = 2.37 \end{aligned}$$

or \$2.37.

Q15.21

In this case, $S_0 = 52$, $K = 50$, $r = 0.12$, $\sigma = 0.30$ and $T = 0.25$.

$$d_1 = \frac{\ln(52/50) + (0.12 + 0.3^2/2)0.25}{0.30\sqrt{0.25}} = 0.5365$$

$$d_2 = d_1 - 0.30\sqrt{0.25} = 0.3865$$

The price of the European call is

$$\begin{aligned} & 52N(0.5365) - 50e^{-0.12 \times 0.25}N(0.3865) \\ &= 52 \times 0.7042 - 50e^{-0.03} \times 0.6504 \\ &= 5.06 \end{aligned}$$

or \$5.06.